



MME 321 Lab 7: Hydraulic System

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3/18/26

1.0 Introduction

The objective of this experiment is to model the translational motion of a piston in a hydraulic system. A hydraulic system is often used in industry to transmit power or high force across distances using incompressible fluids. For the purposes of this lab, MATLAB's Simscape package will be used to create a hydraulic system to translational motion system model. Particularly, the linear actuator element in Simscape is used to model actuators that might be used in the real world for purposes of high precision or high strength. Simscape will be used to view the dynamic output response of the piston's motion in accordance with the given hydraulic system parameter values. This allows engineers to create a predictive solution to a real world problem such as pump specifications.

The hydraulic system is required to power the piston and the resulting motion of the piston will be shown and discussed within the contents of this report.

2.0 Background Theory

Hydraulic System Principles

Hydraulics create motion within the linear actuators by applying the flow energy of the incompressible fluid as well as its pressure along the surface area of the actuator to generate translational motion from hydraulic motion.

Pascal's Law

Pascal's law says that pressure is equal to force per unit area as shown in equation (1). Therefore, by multiplying the pressure of the fluid by the area of the site of contact between the fluid and actuator, we can calculate the force applied to the actuator.

$$P = \frac{F}{A} \quad (1)$$

Hydraulic System Components

Briefly describe the purpose and function of the following components:

- Pump - uses a motor to move fluid from reservoir to rest of the system.
- Reservoir - stores fluid to be drawn away and used in the system.
- Directional Control Valve (4/3 DCV) - blocks or allows fluid to pass based on the requirements of the system at a given time.
- Pressure Relief Valve - reduces pressure of fluid in the case that it becomes dangerously high.
- Check Valve - limits the flow of fluid to only 1 direction to prevent backflow.
- Flow Control Valve - controls flow rate of the fluid.

- Hydraulic Cylinder - generates translational motion from hydraulic pressure and flow energy.

The components work together by ensuring that the proper amount of pressure difference and fluid flow are maintained in order to keep a smoothly working piston cycle with minimized energy loss. Some components such as pressure relief valve also serve to prevent dangerous situations for workers such as a pipe exploding.

3.0 System Modeling in Simulink

Different blocks were used in the simulink model to create the desired system. Their purposes within the system are as follows:

- Hydraulic Reservoir - reference for hydraulic system
- Fixed Displacement Pump - uses DC motor to create flow from the reservoir to the directional control valve
- Check Valve - creates 1 directional flow
- Pressure Relief Valve - regulates the pressure of the flowing fluid
- 4/3 Directional Control Valve - controls the direction of fluid flow with 4 connecting lines and 3 state positions.
- Double-Acting Hydraulic Cylinder - translates flow and pressure energy from fluid into translational motion
- Mass–Spring–Damper System -
- Sensors and Scopes - Sensors allow us to collect data to ensure that the flow rate is correct and stable. The scopes allow us to track the position of the hydraulic cylinder.

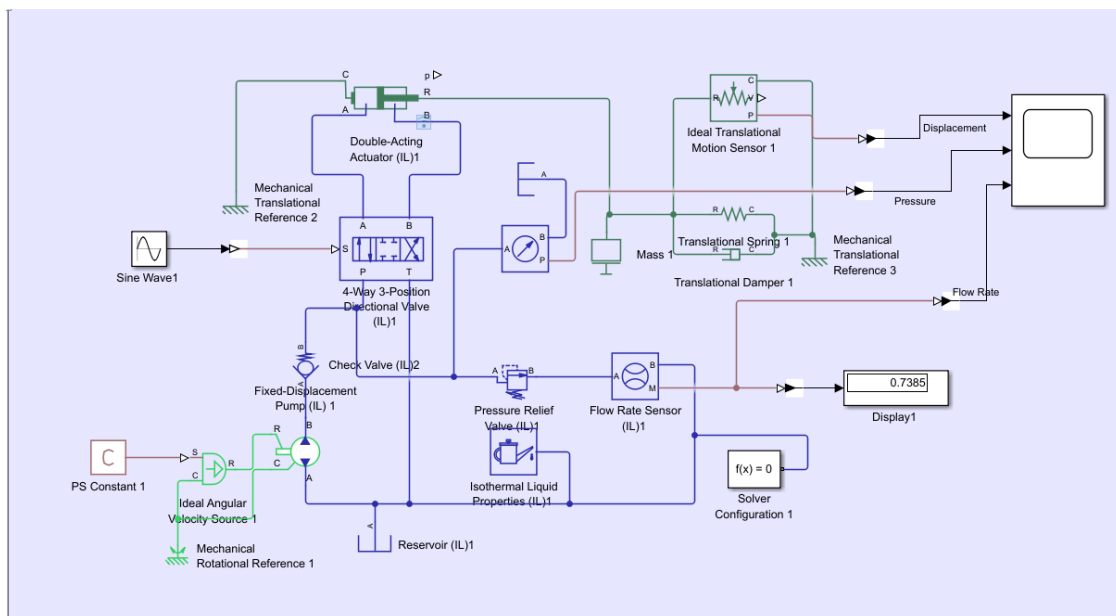


Fig. 1

Fig.1 shows the diagram of the constructed Simscape hydraulic model.

4.0 Simulation Parameters

System parameters include:

- Pump displacement - $30 \text{ cm}^3/\text{rev}$
- Pump rotational speed - 1800 rpm
- Fluid density - 998 kg/m^3
- Fluid bulk modulus
- Pressure relief valve setting - 1 MPa
- Cylinder piston area - 0.01 m^2
- Mass of the load - 1000 kg
- Spring constant - 1000 N/m
- Damping coefficient - $20 \text{ N}\cdot\text{s/m}$

5.0 Simulation Results

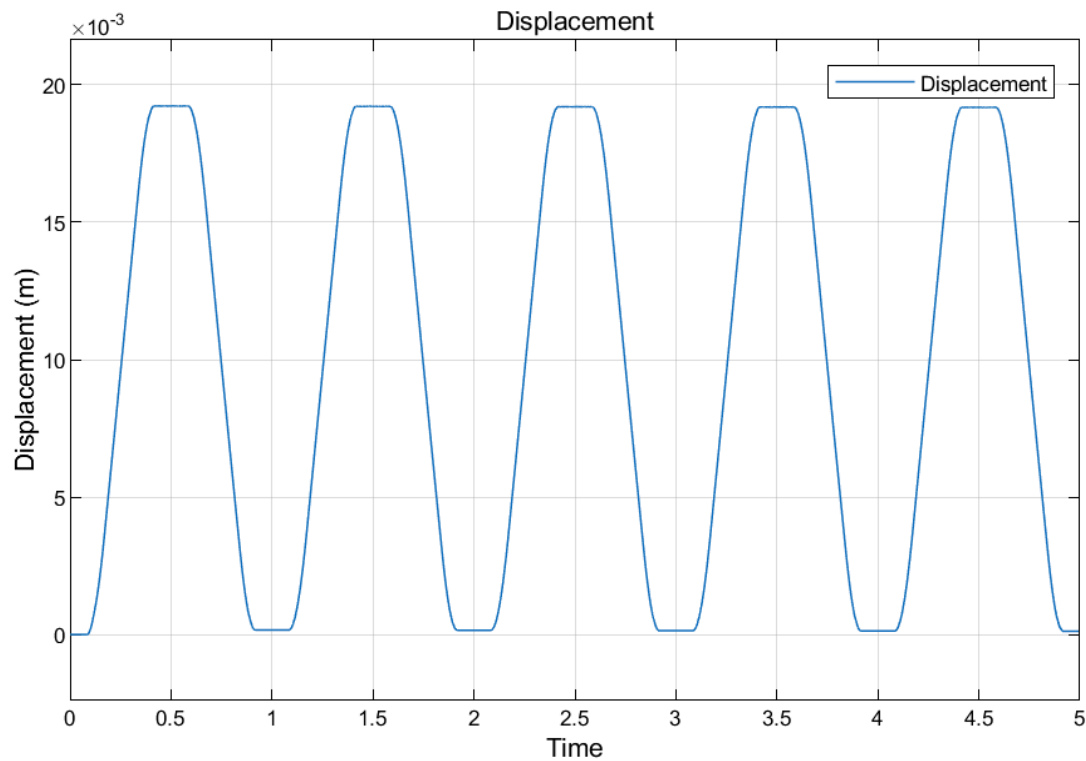


Fig. 2

Fig. 2 shows the cyclical motion of the translational piston caused by the fluid of the system. The piston moves almost 20mm before returning to its original position.

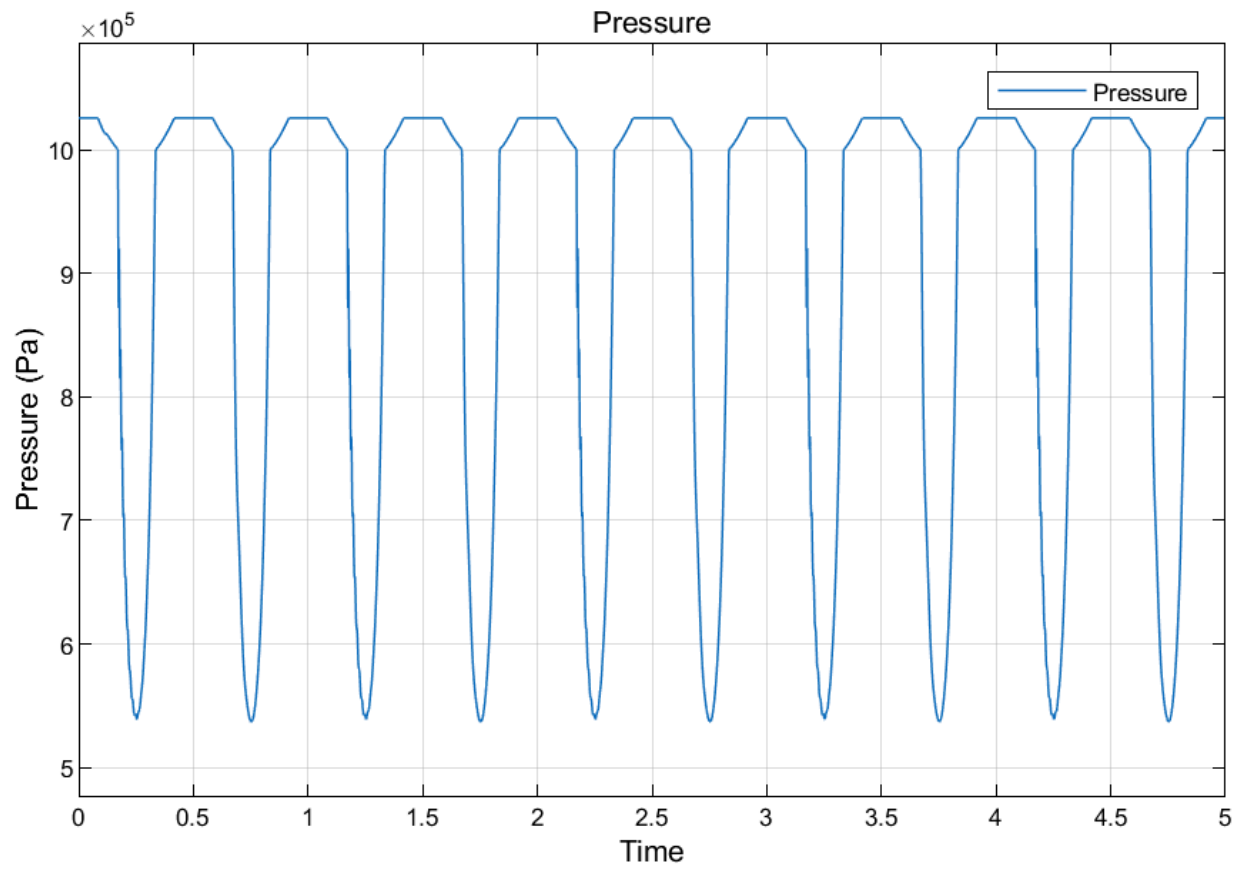


Fig. 3

Fig. 3 shows the pressure of the relief valve as a function of time. The pressure peaks at around 10.3×10^5 Pa.

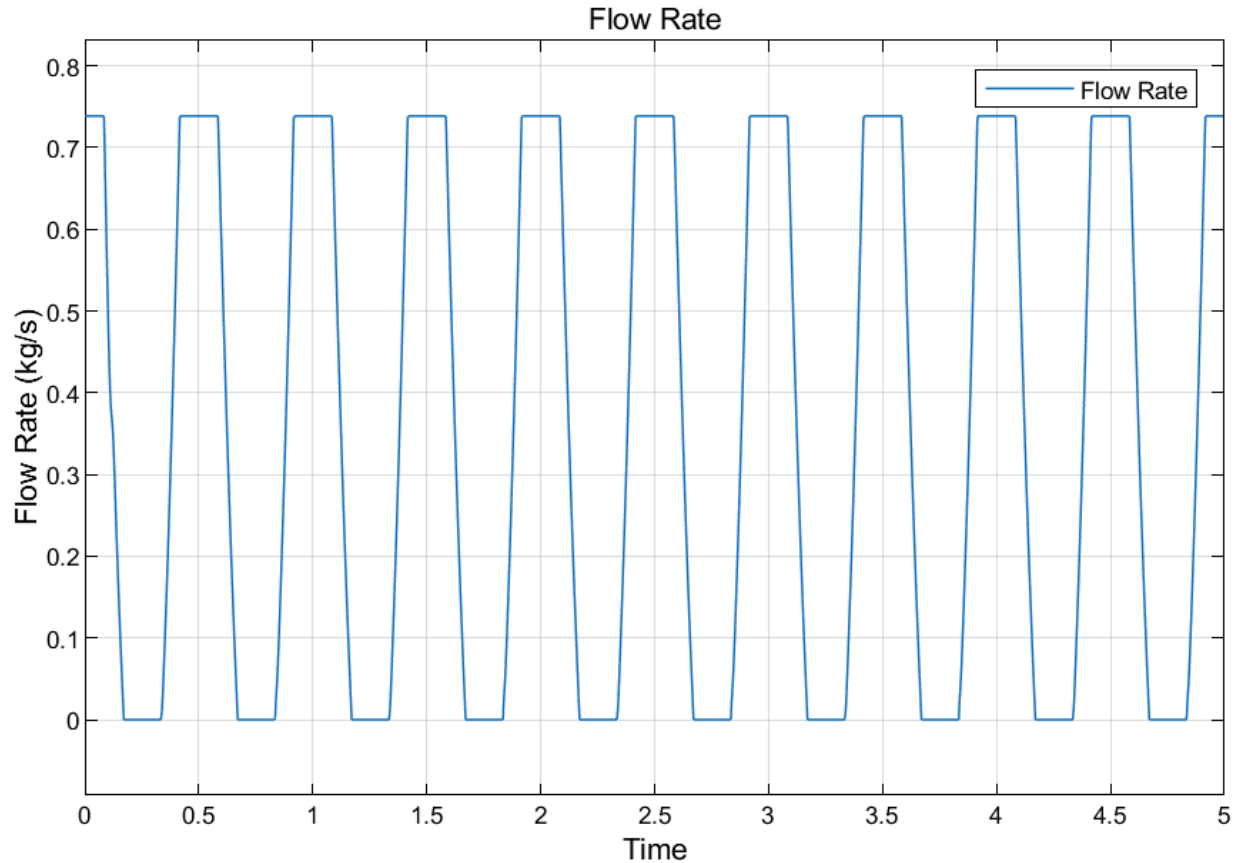


Fig. 4

Fig.4 shows the rate of flow of the fluid through the relief valve. The signal shows that the flow rate is cyclical and has a maximum flow rate of about 0.7385 kg/s.

6.0 Discussion

Overall, the hydraulic Simscape simulation was a success. It allowed the flow rate, pressure, and piston displacement to be calculated for the given system. We would expect the displacement of the linear actuator to be cyclic because it moves back and forth from its original position, so this is correct. Additionally, the pressure and flow rate of the relief valve would be cyclical because the relief valve only functions when it is required to regulate pressure, then it closes.

Without the directional control valve the actuator would cause backflow which would cause misfiring and pressure loss within the system. It is a very necessary part of the system. Much like the pressure relief valve which functions to siphon away fluid when pressure builds up too high. This prevents water from damaging pipes or causing leaks or explosions. Without these two components, the actuator would fail to convert hydraulic pressure and flow energy into usable translational motion energy.

7.0 Conclusion

In this experiment, a hydraulic system was successfully modeled using MATLAB Simscape to analyze the translational motion of a piston driven by hydraulic fluid. The simulation demonstrated how hydraulic energy, using principles such as Pascal's Law, can be effectively converted into translational motion through a translational actuator. By incorporating key system components such as a DC motor driven pump, directional control valve, pressure relief valve, and other elements, we can effectively model the dynamic response of a hydraulic system.